ISS-Box Wiring

Build reference for the Perma-Proto assembly — pin map, net list, modules, passives, and the order to solder them in.

RASPBERRY PI PICO WH RP2040 13 OF 26 GPIO USED 13 NETS

PIN MAP

Laid out as the board physically is: pins 1–20 down the left edge, 21–40 down the right. Free pins are shown deliberately — the gaps between unrelated signals are there so the Perma-Proto Half + Quarter build has room to solder.

5 V — VBUS | 3.3 V | GROUND | SIGNAL | FREE

LEFT EDGE · PINS 1–20

1	GP0	—
2	GP1	—
3	GND	Ground rail
4	GP2	Ring data → 470 Ω
5	GP3	—
6	GP4	I²C SDA
7	GP5	I²C SCL
8	GND	Ground rail
9	GP6	—
10	GP7	Digit data — no resistor
11	GP8	—
12	GP9	Buzzer → 1 k → Q1 base
13	GND	Q1 emitter
14	GP10	—
15	GP11	RTC SQW — 1 Hz
16	GP12	—
17	GP13	Encoder SW
18	GND	Encoder GND
19	GP14	Encoder CLK
20	GP15	Encoder DT

RIGHT EDGE · PINS 21–40

21	GP16	Arcade switch
22	GP17	Arcade lamp +
23	GND	Switch + lamp –
24	GP18	—
25	GP19	—
26	GP20	—
27	GP21	—
28	GND	—
29	GP22	—
30	RUN	—
31	GP26	Gas A0 ← divider · ADC0
32	GP27	Gas EN — low = heater on
33	AGND	Divider + 100 nF ground
34	GP28	—
35	VREF	—
36	3V3	LCD · SCD40 · Encoder +
37	3V3_EN	—
38	GND	Ground rail
39	VSYS	← 1N5817 · box only
40	VBUS	Ring · Digits · Gas VCC

MODULES

NeoPixel ring — 24 PX

DIN	470 Ω → pin 4
5 V	pin 40
GND	pin 38
DOUT	not connected

NeoPixel digits — 1 OF 4 FITTED

DIN	pin 10 direct
Series R	330 Ω already on the module
5 V	pin 40
GND	pin 38
DOUT	→ next digit

LCD1602 RGB — DFR0464

VCC	pin 36 — 3.3 V
GND	pin 38
SDA	pin 6
SCL	pin 7

SCD40 — 0X62

VIN	pin 36 — 3.3 V
GND	pin 38
SDA	pin 6
SCL	pin 7
Decoupling	C4 at VIN — see passives
Peak draw	~205 mA every 5 s
Replaces	SHT3x at 0x44 — same four wires

DS3231 RTC — 0X68

Vin	pin 36 — 3.3 V
GND	pin 38
SDA	pin 6
SCL	pin 7
SQW	pin 15 — optional
32K / RST	not connected
Battery	CR1220, fit before closing up

Rotary encoder — KY-040

+	pin 36 — 3.3 V
GND	pin 18
CLK	pin 19
DT	pin 20
SW	pin 17

Arcade button — ILLUMINATED

Switch	pin 21 · pin 23
Lamp +	pin 22
Lamp –	pin 23

Buzzer — UDB-05B MAGNETIC

+	5 V rail
–	Q1 collector
Q1 base	1 kΩ → pin 12
Q1 emitter	pin 13

USB-C breakout — ADAFRUIT 4090

VBUS	5 V rail + D1 anode
GND	ground rail
CC1/CC2	already resistored on the board
D+/D–	leave unconnected — power only

Gas sensor — MICS-5524

VCC	pin 40 — 5 V
GND	pin 38
EN	pin 32
A0	divider → pin 31

NET LIST

Perfboard is built net by net, not module by module — every point on one row below is the same copper. Tick each row off as it is joined.

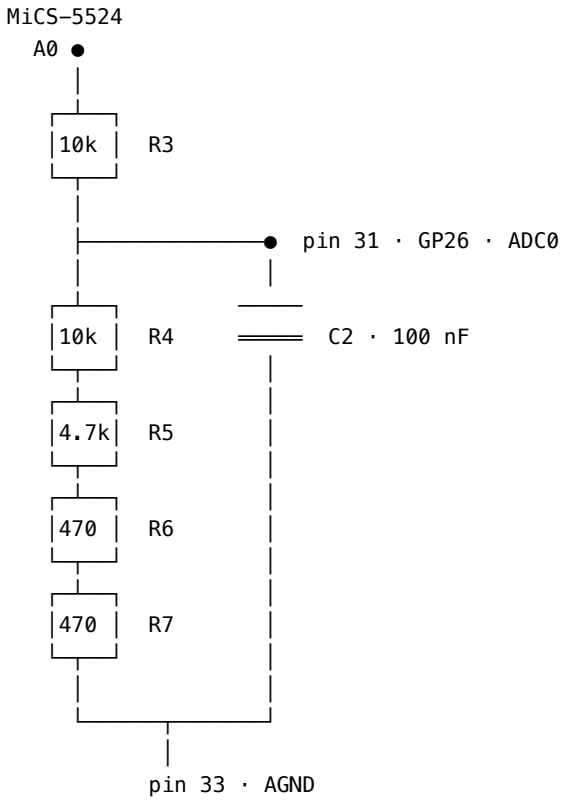
NET	EVERY POINT ON IT
GND	Pico 3, 8, 13, 18, 23, 28, 33, 38 · ring GND · digits GND · LCD GND · SCD40 GND · encoder GND · arcade switch leg · arcade lamp – · gas GND · Q1 emitter · divider foot · C1 – · C2 · C4 · USB-C GND
5 V	USB-C VBUS · ring 5 V · digits 5 V · gas VCC · buzzer + · C1 + · D1 anode *(bench: Pico pin 40 instead of the breakout)*
3V3	Pico 36 · LCD VCC · SCD40 VIN · DS3231 Vin · encoder + · C4
VSYS	Pico 39 · D1 cathode *(banded end)*
SDA	Pico 6 · LCD SDA · SCD40 SDA · DS3231 SDA
SCL	Pico 7 · LCD SCL · SCD40 SCL · DS3231 SCL
SQW	Pico 15 → DS3231 SQW *(internal pull-up, no resistor)*
RING	Pico 4 → R1 → ring DIN *(R1 at the ring end)*
DIGIT	Pico 10 → digit 1 DIN, **no external resistor**; DOUT chains on
BUZZ	Pico 12 → R8 → Q1 base
ENC	Pico 17 → SW · 19 → CLK · 20 → DT
BTN	Pico 21 → switch · 22 → lamp +
GAS	Pico 31 → divider tap + C2 · Pico 32 → EN

> **Nothing else joins the 5 V rail to VSYS or to 3V3.** Those three are separate copper. The only link between 5 V and VSYS is through D1, and 3V3 is made by the Pico itself.

> **Check the SCD40 carrier before soldering it in.** The Sensirion die itself takes 2.4–5.5 V, so 3.3 V suits it — but some AliExpress carriers put a 5 V→3.3 V regulator in front, and feeding one of those 3.3 V lands it below its dropout. Look for a small three-pin regulator next to the input pins. If there is one, the board wants 5 V — and then its pull-ups put 5 V on SDA and SCL, which the RP2040 will not tolerate on a shared bus.

GAS SENSOR DIVIDER

The sensor runs at 5 V and its output swings to 5 V; the ADC tolerates 3.3 V. Sized for the Raspberry Pi adapter's 5.1 V, not a round 5.0 V.



SUPPLY	AT GP26	ADC RANGE	
5.00 V	3.050 V	92.4 %	
5.10 V nominal	3.111 V	94.3 %	Pi adapter
5.36 V worst case	3.270 V	99.1 %	Still under the rail

WHY AN RTC IN A WIFI BOX

The clock was going to come from NTP. The DS3231 changes that, and it is worth being clear about which one is in charge.

> **NTP cannot answer until WiFi associates** — seconds of blank display on every power-up, and nothing at all if the router is down or the box is somewhere else. For a toy whose whole specification is *instant-on and survives a yanked plug*, that is the wrong clock. The DS3231 has the time on the bus immediately and keeps it on a CR1220 with the power off. **NTP still runs — but to correct the RTC, not to be the clock.**

> **It is a TCXO, which is the point.** A plain crystal RTC drifts minutes a month as the room warms and cools. This one holds ±2 ppm — about two minutes a year — so even with the WiFi password wrong all year, the clock stays right.

> **Three sets of pull-ups now share the bus.** LCD, SCD40 and DS3231 each bring their own 10 kΩ, so the bus sees roughly 3.3 kΩ. That is fine — about 1 mA of sink against the 3 mA these parts can take — and for a bit-banged bus the stiffer pull-up actually sharpens the rising edges.

DIGIT DATA LINE

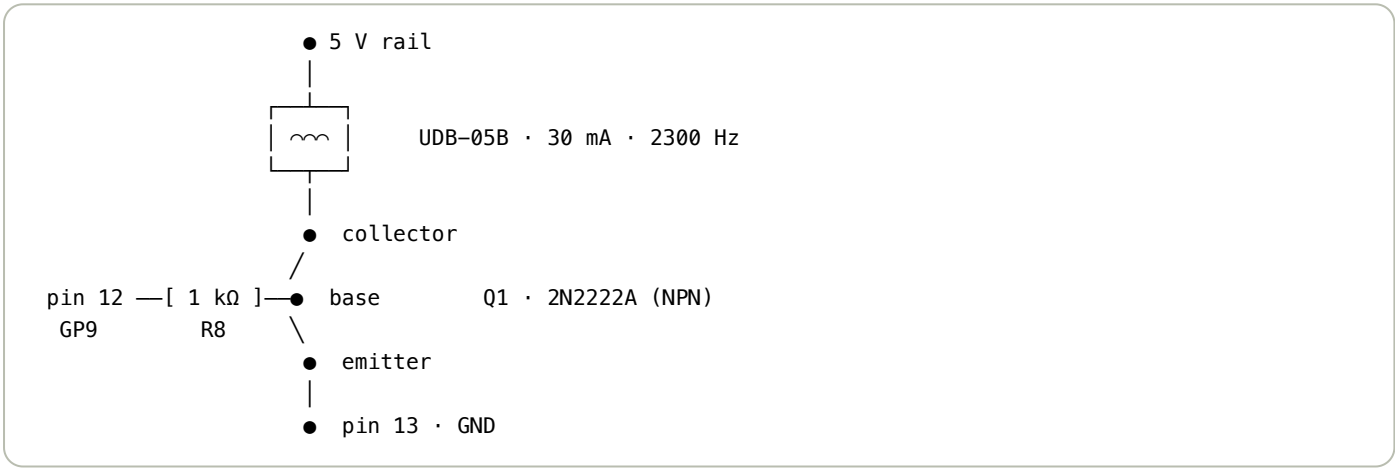
The RGBDigit datasheet, read after the digits were already working, contradicts two things we had settled.

> **No external resistor on the digit data line.** The module already carries 330 Ω in its Di line — the datasheet says so outright: *"an external resistor is not necessary"*. Our R2 would have put **800 Ω** in series, and that is a plausible contributor to the digit fault that metered 470 Ω correctly and still would not pass data. The ring's R1 stays: that module has no resistor of its own.

> **The digit's logic threshold is above what the Pico can drive.** VIH is 0.7 × VDD, so at VDD = 5 V a high must reach **3.5 V** — and the RP2040 drives 3.3 V. On paper the margin is negative. One digit on a short breadboard lead works anyway; four soldered inside a box is where that stops being reliable. The **SN74HCT125N** in the parts box is the fix: power it at 5 V, GP7 into **1A**, **1Y** to digit 1 DIN, **10E** tied low, unused inputs to ground.

BUZZER DRIVER

The UDB-05B is a **magnetic** transducer, not a piezo: 5 V rated, **30 mA**, resonant at **2300 ± 400 Hz**. Both numbers matter.



> **30 mA is over the RP2040's 12 mA per-pin rating.** Driving the buzzer straight from GP9 runs the pin out of spec. The 2N2222A that turned out not to be needed for the arcade lamp goes here instead — and it lets the buzzer run at 5 V, which is where the datasheet's 83 dBA is measured. **No code change:** PWM drives a base perfectly well.

> **Melodies live in A6–F7 (1760–2794 Hz), straddling resonance.** A magnetic buzzer off resonance is quiet *and* current-hungry — its impedance is mostly inductive reactance, which falls with frequency, so low notes draw more current to make less noise. The first version of these melodies sat at 260–1046 Hz, one to three octaves too low.

PASSIVE COMPONENTS

REF	VALUE	WHERE	WHY
R1	470 Ω	Pin 4 → ring DIN	Damps ringing. Mount close to the ring, not the Pico.
R2	~~470 Ω~~ none	Pin 10 → digit DIN	**Do not fit.** The RGBDigit already has 330 Ω in its Di line; an external one makes 800 Ω.
R3	10 kΩ	A0 → pin 31	Divider, upper leg.
R4–R7	10 k + 4.7 k + 470 + 470	Pin 31 → AGND	Divider lower leg in series — 15.64 kΩ total.
C1	1000 µF 6.3 V	5 V ⇔ GND at the pixels	**Polarised.** Stripe to ground. Fit with power off.
C2	100 nF	Pin 31 → AGND	Settles the ADC sampling.
C3+	100 nF	Across each pixel board	Local decoupling. Optional but cheap.
R8	1 kΩ	Pin 12 → Q1 base	Base current limit. Anything 470 Ω–4.7 kΩ works.
Q1	2N2222A	Buzzer low side	NPN. The buzzer's 30 mA is over the pin rating.
D1	1N5817	USB-C VBUS → pin 39	Schottky. Lets the panel port and micro-USB coexist.
C4	100 nF ∥ 10 µF	SCD40 VIN → GND	Sensirion asks for both. It pulses to ~205 mA every 5 s off a 3.3 V rail the LCD also sits on — go to 100 µF if you have one.

POWER

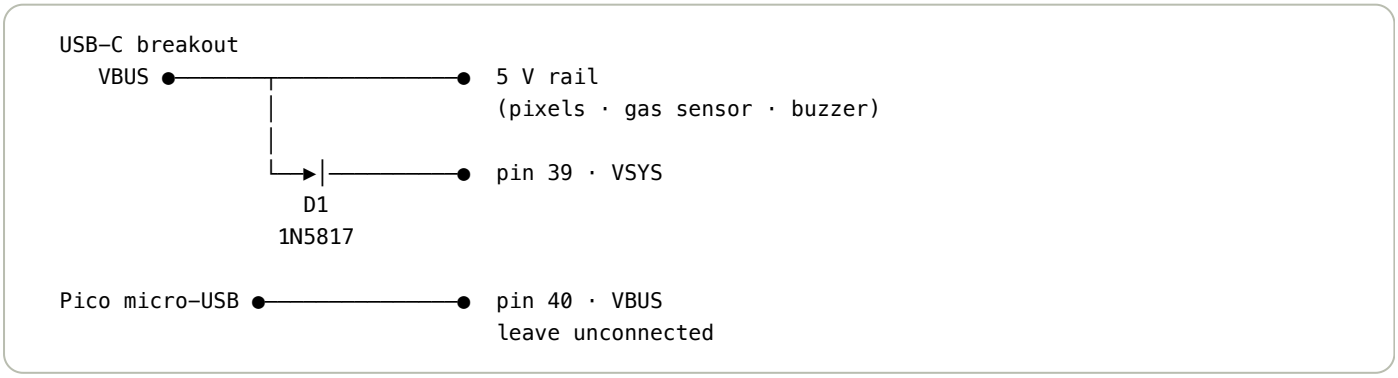
The onboard regulator supplies 3.3 V on pin 36 for the LCD, the SCD40 and the encoder in both arrangements below. What changes is where the 5 V comes from.

On the breadboard now

Everything through the Pico's own micro-USB connector — the Raspberry Pi USB-C adapter via a USB-C-to-micro-USB cable. **VBUS (pin 40) distributes 5 V** to the pixels, the gas sensor and the buzzer. One source, so nothing can back-feed anything.

In the finished box — panel USB-C

The Adafruit 4090 breakout gives the case a proper USB-C hole instead of a micro-USB cable poking out of it. That reintroduces a second supply, which is what the Schottky is for.



The diode is what lets both ports be plugged in at once. VSYS accepts anything from 1.8 to 5.5 V and the Pico's own VBUS reaches VSYS through an identical internal Schottky, so whichever source is higher wins and neither drives current back into the other. That means the box can stay powered from the panel while you reflash it over micro-USB.

Pin 40 must stay unconnected in that arrangement. Tying it to the 5 V rail would wire the two supplies straight together, bypassing the diode entirely — the one thing it exists to prevent.

3.3 V for the encoder, never 5 V. Its onboard pull-ups sit on CLK and DT, so whatever feeds + is what those signal lines idle at. The RP2040 is not 5 V tolerant.

The LCD also stays on 3.3 V for the same reason — its pull-ups share the I²C bus with the Pico's pins.

SOLDER ORDER

Rails first, then one module at a time, powering up between each. A fault found after two joints is a fault you can see; found after forty, it is a hunt.

#	FIT	THEN CHECK
1	Pico headers / socket	It seats and sits flat. Nothing else on the board yet.
2	Ground rail, every GND point	Continuity from Pico pin 3 to every other GND point.
3	USB-C breakout, D1, C1	**Meter before anything is powered:** 5 V rail to GND must *not* be a short. Then power the panel port alone — 5 V on the rail, ~4.8 V on VSYS.
4	3V3 from pin 36	3.3 V present. The Pico should now boot from the panel port on its own.
5	I²C pair + LCD + SCD40 + DS3231 + C4	Flash and watch the scan: `0x3E`, `0x60`, `0x62`, `0x68`.
6	Ring, with R1 at the ring	Comet animation runs.
7	Digits, straight to pin 10	All four digits now — then raise `DIGITS_FITTED` to 4.
8	Q1, R8, buzzer	Startup tune. Loud, if the transistor is the right way round.
9	Encoder, arcade button + lamp	Detents count the right way; lamp breathes.
10	Gas divider, C2, MiCS-5524	Last because it is the fiddliest — four resistors nose to tail. Baseline settles around 0.1–0.4 V.

The one joint that has already bitten you: the 470 Ω to the digit DIN. It read 470 Ω on the meter and still would not pass data, because a probe tip pressed onto a pad makes a connection the solder does not. Tug-test that wire, and check continuity from the far end of the resistor to the digit pad itself — not to the pad it is supposed to be joined to.

THINGS THAT READ BACKWARDS

ITEM	WHAT TO EXPECT
Gas `EN`	HIGH *disables* the heater. Driven low.
Encoder CLK/DT	Must be consecutive GPIO, CLK first — the PIO decoder reads them as a two-bit field.
Encoder PIO	Runs on PIO1. Its program needs instruction address 0, which WS2812 holds on PIO0.
I²C	Bit-banged in software. The hardware block never started a transfer against this LCD.
Digit pixels	8 per digit: offsets 0–6 are segments a–g, offset 7 is the dot.
Buzzer notes	Low notes are quieter *and* draw more current. It is magnetic, resonant at 2300 Hz.
The transistor	It is on the buzzer, not the illuminated button — the opposite way round to the guess.
RTC `SQW`	Open drain. It needs a pull-up to read as anything; GP11's internal one is enough.
RTC registers	BCD, not binary. And the address pointer wraps 12h → 00h mid-read.